DC1 universal board: population summary

One 100×100 mm board, four builds (a–d). Each build fits only the parts listed for its letter; every other resistor footprint is left open. Never fit a 0 Ω link in a DNP position.

Card	Neurons	Resistors fitted	5% (weights, bias)	1% (balance, feedback)	DNP footprints
DC1a	N0–N3	191	183	8	229
DC1b	N4–N7	230	222	8	190
DC1c	N8–N11	181	173	8	239
DC1d	N12–N15	141	133	8	279

Fixed parts (fit on every card)

Ref	Part	Cards
U1	TL074 quad op-amp, SOIC-14	all cards
C_U1P, C_U1N	0.1 µF 50 V 0805 decoupling	all cards
J_STRADDLE_EDGE	76-pin edge straddle	all cards

value = solder 5% 0805. **value** with **1%** tag on blue = solder 1% precision part. **DNP** = leave pads OPEN (never 0Ω). – on grey = no footprint on the board.

Board layout: IN0–IN31 bank on the left half, IN32–IN63 on the right, one row per input, eight columns per bank (POS/INV for local N0–N3). The 16 central resistors sit in two columns at x = 47 / 52.6 mm, y = 59–94 mm. Reference numbers in each cell match the silkscreen.

DC1a · weights IN0–IN31 (left bank) · neurons N0–N3

value = solder 5% 0805. **value** with **1%** tag on blue = solder 1% precision part. **DNP** = leave pads OPEN (never 0Ω). – on grey = no footprint on the board.

IN	POS N0 (POS_N0)	INV N0 (INV_N0)	POS N1 (POS_N1)	INV N1 (INV_N1)	POS N2 (POS_N2)	INV N2 (INV_N2)	POS N3 (POS_N3)	INV N3 (INV_N3)
IN0	R001 DNP	R002 30k	R003 DNP	R004 DNP	R005 DNP	R006 DNP	R007 DNP	R008 47k
IN1	R009 390k	R010 DNP	–	R011 DNP	–	R012 110k	R013 2M	R014 DNP
IN2	R015 160k	R016 DNP	–	R017 DNP	R018 1.5M	R019 DNP	R020 120k	–
IN3	R021 180k	R022 DNP	R023 DNP	R024 DNP	R025 DNP	R026 680k	R027 160k	–
IN4	R028 DNP	R029 130k	R030 DNP	R031 DNP	–	R032 510k	R033 150k	–
IN5	R034 DNP	R035 360k	R036 DNP	R037 DNP	R038 110k	R039 DNP	R040 240k	–
IN6	R041 DNP	R042 270k	–	R043 DNP	R044 200k	R045 DNP	R046 430k	–
IN7	–	R047 DNP	–	R048 DNP	R049 DNP	R050 150k	R051 DNP	R052 180k
IN8	R053 360k	R054 DNP	R055 DNP	R056 82k	–	R057 DNP	R058 DNP	R059 130k
IN9	R060 200k	R061 DNP	R062 360k	R063 DNP	R064 240k	R065 DNP	R066 470k	–
IN10	R067 390k	R068 DNP	–	R069 DNP	–	R070 560k	–	R071 620k
IN11	R072 DNP	R073 DNP	R074 DNP	R075 DNP	R076 680k	R077 DNP	–	–
IN12	R078 DNP	R079 160k	R080 DNP	–	–	R081 DNP	–	R082 DNP
IN13	R083 DNP	R084 220k	–	R085 DNP	–	R086 360k	R087 3.3M	R088 DNP
IN14	R089 DNP	R090 270k	R091 DNP	R092 DNP	–	R093 330k	–	R094 1.1M
IN15	–	R095 150k	–	R096 DNP	–	R097 DNP	–	R098 300k
IN16	R099 DNP	R100 DNP	R101 DNP	R102 DNP	R103 DNP	R104 160k	R105 DNP	R106 47k
IN17	R107 DNP	R108 DNP	R109 DNP	R110 DNP	R111 DNP	R112 220k	R113 DNP	R114 56k
IN18	R115 DNP	R116 910k	–	R117 DNP	R118 DNP	R119 160k	R120 DNP	R121 91k
IN19	R122 82k	–	R123 DNP	R124 DNP	R125 DNP	R126 220k	–	R127 130k
IN20	R128 430k	–	–	R129 DNP	R130 DNP	R131 180k	–	R132 2.2M
IN21	–	R133 120k	–	R134 DNP	R135 DNP	R136 100k	–	R137 430k
IN22	–	R138 130k	–	R139 DNP	R140 DNP	R141 91k	–	R142 2M
IN23	R143 DNP	R144 180k	–	R145 DNP	R146 DNP	R147 47k	–	R148 560k
IN24	–	R149 150k	R150 DNP	R151 DNP	R152 DNP	R153 240k	R154 75k	R155 DNP
IN25	R156 430k	R157 DNP	R158 DNP	R159 DNP	R160 DNP	R161 DNP	R162 75k	R163 DNP
IN26	R164 390k	R165 DNP	–	R166 DNP	–	R167 270k	R168 150k	R169 DNP
IN27	R170 160k	R171 DNP	–	R172 DNP	–	R173 330k	R174 150k	R175 DNP
IN28	R176 DNP	R177 300k	–	R178 DNP	R179 DNP	R180 680k	R181 620k	–
IN29	R182 DNP	R183 240k	–	R184 DNP	R185 DNP	R186 200k	R187 330k	–
IN30	–	R188 1.6M	R189 DNP	R190 DNP	R191 DNP	R192 330k	R193 560k	–
IN31	R194 620k	R195 DNP	R196 DNP	R197 75k	R198 220k	–	R199 180k	–

DC1a · weights IN32–IN63 (right bank) · neurons N0–N3

value = solder 5% 0805. **value** with **1%** tag on blue = solder 1% precision part. **DNP** = leave pads OPEN (never 0Ω). – on grey = no footprint on the board.

IN	POS N0 (POS_N0)	INV N0 (INV_N0)	POS N1 (POS_N1)	INV N1 (INV_N1)	POS N2 (POS_N2)	INV N2 (INV_N2)	POS N3 (POS_N3)	INV N3 (INV_N3)
IN32	R200 **360k**	–	R201 DNP	R202 **91k**	R203 **910k**	R204 DNP	R205 **27k**	R206 DNP
IN33	R207 **1M**	R208 DNP	R209 **110k**	–	R210 **180k**	R211 DNP	R212 **91k**	–
IN34	R213 DNP	R214 **1.3M**	R215 DNP	–	R216 **180k**	R217 DNP	R218 **390k**	R219 DNP
IN35	R220 DNP	R221 **680k**	R222 DNP	R223 **200k**	R224 **220k**	R225 DNP	–	R226 **330k**
IN36	R227 DNP	R228 **470k**	–	R229 DNP	R230 **270k**	R231 DNP	R232 DNP	R233 DNP
IN37	R234 DNP	R235 **1.2M**	R236 DNP	R237 **470k**	R238 DNP	R239 **360k**	–	R240 DNP
IN38	R241 **330k**	–	R242 DNP	R243 DNP	R244 DNP	R245 **430k**	R246 **300k**	–
IN39	R247 **110k**	R248 DNP	R249 DNP	R250 DNP	R251 DNP	R252 **75k**	R253 DNP	R254 **150k**
IN40	R255 DNP	R256 **510k**	R257 DNP	–	R258 DNP	R259 DNP	–	R260 **43k**
IN41	R261 **330k**	R262 DNP	R263 **110k**	–	R264 DNP	–	–	R265 **240k**
IN42	R266 **390k**	R267 DNP	R268 DNP	–	R269 **150k**	–	R270 DNP	–
IN43	R271 **390k**	R272 DNP	R273 DNP	–	R274 **91k**	R275 DNP	–	R276 **750k**
IN44	R277 **1.2M**	R278 DNP	R279 DNP	–	R280 **330k**	R281 DNP	–	R282 **2.4M**
IN45	R283 DNP	R284 DNP	R285 **390k**	R286 DNP	R287 DNP	R288 DNP	–	R289 **2M**
IN46	R290 DNP	–	R291 DNP	–	R292 DNP	R293 **300k**	R294 DNP	R295 **1M**
IN47	R296 **68k**	R297 DNP	R298 DNP	R299 DNP	R300 DNP	R301 **110k**	R302 DNP	R303 **110k**
IN48	R304 DNP	R305 DNP	–	R306 DNP	R307 DNP	R308 **240k**	R309 **470k**	–
IN49	R310 **240k**	R311 DNP	R312 **110k**	–	R313 **270k**	–	–	R314 **470k**
IN50	R315 **330k**	R316 DNP	–	–	R317 DNP	R318 **680k**	–	R319 **910k**
IN51	R320 **560k**	R321 DNP	R322 DNP	R323 **680k**	R324 DNP	R325 **330k**	R326 **470k**	R327 DNP
IN52	R328 **560k**	R329 DNP	R330 DNP	R331 **1.2M**	R332 **3.3M**	–	R333 DNP	R334 DNP
IN53	R335 **560k**	R336 DNP	R337 **200k**	R338 DNP	R339 **390k**	R340 DNP	–	R341 **1.1M**
IN54	–	R342 **430k**	–	R343 DNP	R344 **270k**	R345 DNP	R346 **430k**	–
IN55	R347 DNP	R348 **82k**	R349 DNP	R350 DNP	R351 DNP	R352 **200k**	R353 DNP	R354 **91k**
IN56	–	R355 **62k**	R356 DNP	R357 DNP	R358 DNP	R359 DNP	–	R360 **75k**
IN57	R361 DNP	R362 DNP	R363 DNP	R364 DNP	R365 **220k**	R366 DNP	–	R367 **470k**
IN58	–	R368 DNP	R369 DNP	R370 DNP	–	R371 **390k**	R372 DNP	R373 DNP
IN59	–	R374 **360k**	R375 DNP	–	–	R376 **200k**	R377 **200k**	R378 DNP
IN60	–	R379 **1.1M**	R380 DNP	R381 **130k**	R382 DNP	R383 **220k**	R384 **330k**	R385 DNP
IN61	R386 **240k**	R387 DNP	R388 DNP	R389 DNP	R390 **130k**	R391 DNP	–	R392 **330k**
IN62	–	R393 **200k**	R394 DNP	R395 **120k**	R396 **100k**	–	–	R397 **110k**
IN63	–	R398 DNP	R399 DNP	R400 DNP	R401 **56k**	R402 DNP	R403 **100k**	R404 DNP

DC1a · central resistors (feedback, bias, balance)

x = 47 / 52.6 mm, y = 59–94 mm. Balance and feedback are 1% (blue); bias is 5%. Red rows stay open.

Ref	X mm	Y mm	Pad 1	Pad 2	Role	Neuron	Value	Tol	Original ref
R405	47	59	+5V_REF	INV_N0	bias	N0	**6.2M**	5%	Rb_N0
R406	52.6	59	+5V_REF	INV_N1	bias	N1	DNP (open)		
R407	47	64	+5V_REF	INV_N2	bias	N2	**1.1M**	5%	Rb_N2
R408	52.6	64	+5V_REF	INV_N3	bias	N3	DNP (open)		
R409	47	69	+5V_REF	POS_N2	bias	N2	DNP (open)		
R410	52.6	69	GND	INV_N1	balance	N1	DNP (open)		
R411	47	74	GND	INV_N2	balance	N2	DNP (open)		
R412	52.6	74	GND	INV_N3	balance	N3	DNP (open)		
R413	47	79	GND	POS_N0	balance	N0	**18.2k**	1%	R_BAL_1_0
R414	52.6	79	GND	POS_N1	balance	N1	**29.4k**	1%	R_BAL_1_1
R415	47	84	GND	POS_N2	balance	N2	**14.7k**	1%	R_BAL_1_2
R416	52.6	84	GND	POS_N3	balance	N3	**22.6k**	1%	R_BAL_1_3
R417	47	89	INV_N0	V_SUM_0	feedback	N0	**100k**	1%	Rf_N0
R418	52.6	89	INV_N1	V_SUM_1	feedback	N1	**100k**	1%	Rf_N1
R419	47	94	INV_N2	V_SUM_2	feedback	N2	**100k**	1%	Rf_N2
R420	52.6	94	INV_N3	V_SUM_3	feedback	N3	**100k**	1%	Rf_N3

Also fit the fixed parts: U1 TL074, C_U1P and C_U1N 0.1 µF 50 V 0805, J_STRADDLE_EDGE.

DC1b · weights IN0–IN31 (left bank) · neurons N4–N7

value = solder 5% 0805. **value** with **1%** tag on blue = solder 1% precision part. **DNP** = leave pads OPEN (never 0Ω). – on grey = no footprint on the board.

IN	POS N4 (POS_N0)	INV N4 (INV_N0)	POS N5 (POS_N1)	INV N5 (INV_N1)	POS N6 (POS_N2)	INV N6 (INV_N2)	POS N7 (POS_N3)	INV N7 (INV_N3)
IN0	R001 DNP	R002 51k	R003 130k	R004 DNP	R005 DNP	R006 75k	R007 75k	R008 DNP
IN1	R009 DNP	R010 200k	–	R011 150k	–	R012 240k	R013 91k	R014 DNP
IN2	R015 DNP	R016 1.3M	–	R017 200k	R018 DNP	R019 DNP	R020 330k	–
IN3	R021 430k	R022 DNP	R023 DNP	R024 150k	R025 680k	R026 DNP	R027 330k	–
IN4	R028 DNP	R029 270k	R030 DNP	R031 150k	–	R032 110k	R033 110k	–
IN5	R034 DNP	R035 510k	R036 DNP	R037 300k	R038 DNP	R039 82k	R040 130k	–
IN6	R041 DNP	R042 100k	–	R043 160k	R044 DNP	R045 56k	R046 82k	–
IN7	–	R047 DNP	–	R048 33k	R049 DNP	R050 47k	R051 390k	R052 DNP
IN8	R053 DNP	R054 100k	R055 470k	R056 DNP	–	R057 110k	R058 150k	R059 DNP
IN9	R060 270k	R061 DNP	R062 DNP	R063 DNP	R064 DNP	R065 1M	R066 DNP	–
IN10	R067 DNP	R068 1.8M	–	R069 160k	–	R070 1.6M	–	R071 DNP
IN11	R072 DNP	R073 330k	R074 DNP	R075 300k	R076 DNP	R077 430k	–	–
IN12	R078 DNP	R079 820k	R080 DNP	–	–	R081 270k	–	R082 1.1M
IN13	R083 DNP	R084 DNP	–	R085 DNP	–	R086 200k	R087 DNP	R088 300k
IN14	R089 510k	R090 DNP	R091 1.8M	R092 DNP	–	R093 130k	–	R094 360k
IN15	–	R095 160k	–	R096 160k	–	R097 33k	–	R098 75k
IN16	R099 DNP	R100 DNP	R101 DNP	R102 220k	R103 220k	R104 DNP	R105 510k	R106 DNP
IN17	R107 DNP	R108 DNP	R109 DNP	R110 330k	R111 DNP	R112 DNP	R113 560k	R114 DNP
IN18	R115 DNP	R116 330k	–	R117 DNP	R118 DNP	R119 DNP	R120 910k	R121 DNP
IN19	R122 DNP	–	R123 DNP	R124 360k	R125 160k	R126 DNP	–	R127 DNP
IN20	R128 240k	–	–	R129 430k	R130 82k	R131 DNP	–	R132 DNP
IN21	–	R133 820k	–	R134 DNP	R135 75k	R136 DNP	–	R137 270k
IN22	–	R138 680k	–	R139 470k	R140 56k	R141 DNP	–	R142 75k
IN23	R143 DNP	R144 DNP	–	R145 91k	R146 51k	R147 DNP	–	R148 24k
IN24	–	R149 91k	R150 300k	R151 DNP	R152 DNP	R153 DNP	R154 DNP	R155 110k
IN25	R156 DNP	R157 300k	R158 360k	R159 DNP	R160 DNP	R161 DNP	R162 DNP	R163 510k
IN26	R164 DNP	R165 750k	–	R166 510k	–	R167 DNP	R168 DNP	R169 1.5M
IN27	R170 160k	R171 DNP	–	R172 120k	–	R173 DNP	R174 DNP	R175 2M
IN28	R176 220k	R177 DNP	–	R178 180k	R179 DNP	R180 200k	R181 220k	–
IN29	R182 300k	R183 DNP	–	R184 1.3M	R185 DNP	R186 330k	R187 62k	–
IN30	–	R188 220k	R189 430k	R190 DNP	R191 150k	R192 DNP	R193 20k	–
IN31	R194 DNP	R195 82k	R196 100k	R197 DNP	R198 68k	–	R199 43k	–

DC1b · weights IN32–IN63 (right bank) · neurons N4–N7

value = solder 5% 0805. **value** with **1%** tag on blue = solder 1% precision part. **DNP** = leave pads OPEN (never 0Ω). – on grey = no footprint on the board.

IN	POS N4 (POS_N0)	INV N4 (INV_N0)	POS N5 (POS_N1)	INV N5 (INV_N1)	POS N6 (POS_N2)	INV N6 (INV_N2)	POS N7 (POS_N3)	INV N7 (INV_N3)
IN32	R200 82k	–	R201 91k	R202 DNP	R203 560k	R204 DNP	R205 DNP	R206 160k
IN33	R207 360k	R208 DNP	R209 91k	–	R210 120k	R211 DNP	R212 680k	–
IN34	R213 390k	R214 DNP	R215 180k	–	R216 680k	R217 DNP	R218 DNP	R219 1.1M
IN35	R220 200k	R221 DNP	R222 DNP	R223 300k	R224 DNP	R225 390k	–	R226 430k
IN36	R227 150k	R228 DNP	–	R229 470k	R230 DNP	R231 560k	R232 DNP	R233 750k
IN37	R234 470k	R235 DNP	R236 620k	R237 DNP	R238 DNP	R239 1M	–	R240 270k
IN38	R241 130k	–	R242 330k	R243 DNP	R244 DNP	R245 330k	R246 220k	–
IN39	R247 330k	R248 DNP	R249 DNP	R250 300k	R251 DNP	R252 68k	R253 27k	R254 DNP
IN40	R255 51k	R256 DNP	R257 120k	–	R258 DNP	R259 110k	–	R260 110k
IN41	R261 51k	R262 DNP	R263 150k	–	R264 470k	–	–	R265 DNP
IN42	R266 82k	R267 DNP	R268 82k	–	R269 470k	–	R270 510k	–
IN43	R271 100k	R272 DNP	R273 110k	–	R274 330k	R275 DNP	–	R276 680k
IN44	R277 390k	R278 DNP	R279 180k	–	R280 510k	R281 DNP	–	R282 300k
IN45	R283 300k	R284 DNP	R285 240k	R286 DNP	R287 DNP	R288 240k	–	R289 510k
IN46	R290 200k	–	R291 820k	–	R292 DNP	R293 300k	R294 680k	R295 DNP
IN47	R296 DNP	R297 360k	R298 DNP	R299 82k	R300 DNP	R301 200k	R302 82k	R303 DNP
IN48	R304 DNP	R305 91k	–	R306 150k	R307 510k	R308 DNP	R309 DNP	–
IN49	R310 DNP	R311 200k	R312 DNP	–	R313 510k	–	–	R314 1.1M
IN50	R315 DNP	R316 180k	–	–	R317 DNP	R318 470k	–	R319 DNP
IN51	R320 DNP	R321 120k	R322 200k	R323 DNP	R324 DNP	R325 2M	R326 DNP	R327 2.2M
IN52	R328 DNP	R329 200k	R330 750k	R331 DNP	R332 DNP	–	R333 DNP	R334 750k
IN53	R335 DNP	R336 360k	R337 DNP	R338 DNP	R339 300k	R340 DNP	–	R341 DNP
IN54	–	R342 130k	–	R343 510k	R344 DNP	R345 470k	R346 DNP	–
IN55	R347 DNP	R348 150k	R349 DNP	R350 160k	R351 DNP	R352 DNP	R353 91k	R354 DNP
IN56	–	R355 DNP	R356 DNP	R357 39k	R358 160k	R359 DNP	–	R360 DNP
IN57	R361 DNP	R362 200k	R363 DNP	R364 1.5M	R365 560k	R366 DNP	–	R367 160k
IN58	–	R368 180k	R369 DNP	R370 270k	–	R371 1.8M	R372 DNP	R373 820k
IN59	–	R374 330k	R375 750k	–	–	R376 680k	R377 DNP	R378 240k
IN60	–	R379 200k	R380 1.6M	R381 DNP	R382 DNP	R383 200k	R384 DNP	R385 560k
IN61	R386 DNP	R387 120k	R388 DNP	R389 330k	R390 DNP	R391 270k	–	R392 DNP
IN62	–	R393 DNP	R394 1M	R395 DNP	R396 200k	–	–	R397 180k
IN63	–	R398 62k	R399 DNP	R400 75k	R401 DNP	R402 DNP	R403 DNP	R404 51k

DC1b · central resistors (feedback, bias, balance)

x = 47 / 52.6 mm, y = 59–94 mm. Balance and feedback are 1% (blue); bias is 5%. Red rows stay open.

Ref	X mm	Y mm	Pad 1	Pad 2	Role	Neuron	Value	Tol	Original ref
R405	47	59	+5V_REF	INV_N0	bias	N4	**330k**	5%	Rb_N4
R406	52.6	59	+5V_REF	INV_N1	bias	N5	**330k**	5%	Rb_N5
R407	47	64	+5V_REF	INV_N2	bias	N6	DNP (open)		
R408	52.6	64	+5V_REF	INV_N3	bias	N7	**390k**	5%	Rb_N7
R409	47	69	+5V_REF	POS_N2	bias	N6	**1.1M**	5%	Rb_N6
R410	52.6	69	GND	INV_N1	balance	N5	DNP (open)		
R411	47	74	GND	INV_N2	balance	N6	DNP (open)		
R412	52.6	74	GND	INV_N3	balance	N7	**16.2k**	1%	R_BAL_1_7
R413	47	79	GND	POS_N0	balance	N4	**14.3k**	1%	R_BAL_1_4
R414	52.6	79	GND	POS_N1	balance	N5	**10k**	1%	R_BAL_1_5
R415	47	84	GND	POS_N2	balance	N6	**13k**	1%	R_BAL_1_6
R416	52.6	84	GND	POS_N3	balance	N7	DNP (open)		
R417	47	89	INV_N0	V_SUM_0	feedback	N4	**100k**	1%	Rf_N4
R418	52.6	89	INV_N1	V_SUM_1	feedback	N5	**100k**	1%	Rf_N5
R419	47	94	INV_N2	V_SUM_2	feedback	N6	**100k**	1%	Rf_N6
R420	52.6	94	INV_N3	V_SUM_3	feedback	N7	**100k**	1%	Rf_N7

Also fit the fixed parts: U1 TL074, C_U1P and C_U1N 0.1 µF 50 V 0805, J_STRADDLE_EDGE.

DC1c · weights IN0–IN31 (left bank) · neurons N8–N11

value = solder 5% 0805. **value** with **1%** tag on blue = solder 1% precision part. **DNP** = leave pads OPEN (never 0Ω). – on grey = no footprint on the board.

IN	POS N8 (POS_N0)	INV N8 (INV_N0)	POS N9 (POS_N1)	INV N9 (INV_N1)	POS N10 (POS_N2)	INV N10 (INV_N2)	POS N11 (POS_N3)	INV N11 (INV_N3)
IN0	R001 DNP	R002 DNP	R003 DNP	R004 51k	R005 DNP	R006 DNP	R007 DNP	R008 DNP
IN1	R009 DNP	R010 180k	–	R011 33k	–	R012 43k	R013 DNP	R014 DNP
IN2	R015 DNP	R016 270k	–	R017 DNP	R018 DNP	R019 62k	R020 DNP	–
IN3	R021 DNP	R022 1M	R023 62k	R024 DNP	R025 DNP	R026 120k	R027 DNP	–
IN4	R028 DNP	R029 180k	R030 43k	R031 DNP	–	R032 160k	R033 DNP	–
IN5	R034 DNP	R035 360k	R036 75k	R037 DNP	R038 DNP	R039 240k	R040 DNP	–
IN6	R041 DNP	R042 330k	–	R043 110k	R044 DNP	R045 DNP	R046 DNP	–
IN7	–	R047 47k	–	R048 39k	R049 27k	R050 DNP	R051 DNP	R052 DNP
IN8	R053 DNP	R054 75k	R055 DNP	R056 91k	–	R057 39k	R058 DNP	R059 DNP
IN9	R060 DNP	R061 620k	R062 DNP	R063 360k	R064 DNP	R065 100k	R066 DNP	–
IN10	R067 DNP	R068 220k	–	R069 240k	–	R070 91k	–	R071 DNP
IN11	R072 DNP	R073 360k	R074 180k	R075 DNP	R076 DNP	R077 110k	–	–
IN12	R078 DNP	R079 430k	R080 300k	–	–	R081 330k	–	R082 DNP
IN13	R083 DNP	R084 220k	–	R085 180k	–	R086 560k	R087 DNP	R088 DNP
IN14	R089 DNP	R090 510k	R091 DNP	R092 160k	–	R093 DNP	–	R094 DNP
IN15	–	R095 68k	–	R096 110k	–	R097 82k	–	R098 DNP
IN16	R099 DNP	R100 110k	R101 91k	R102 DNP	R103 47k	R104 DNP	R105 DNP	R106 DNP
IN17	R107 DNP	R108 160k	R109 220k	R110 DNP	R111 47k	R112 DNP	R113 DNP	R114 DNP
IN18	R115 DNP	R116 120k	–	R117 160k	R118 62k	R119 DNP	R120 DNP	R121 DNP
IN19	R122 DNP	–	R123 330k	R124 DNP	R125 75k	R126 DNP	–	R127 DNP
IN20	R128 470k	–	–	R129 390k	R130 DNP	R131 DNP	–	R132 DNP
IN21	–	R133 130k	–	R134 240k	R135 DNP	R136 910k	–	R137 DNP
IN22	–	R138 120k	–	R139 DNP	R140 DNP	R141 270k	–	R142 DNP
IN23	R143 DNP	R144 130k	–	R145 DNP	R146 DNP	R147 180k	–	R148 DNP
IN24	–	R149 200k	R150 DNP	R151 130k	R152 56k	R153 DNP	R154 750k	R155 DNP
IN25	R156 DNP	R157 470k	R158 DNP	R159 150k	R160 750k	R161 DNP	R162 DNP	R163 DNP
IN26	R164 DNP	R165 560k	–	R166 270k	–	R167 DNP	R168 DNP	R169 2.2M
IN27	R170 470k	R171 DNP	–	R172 DNP	–	R173 820k	R174 DNP	R175 DNP
IN28	R176 DNP	R177 DNP	–	R178 240k	R179 270k	R180 DNP	R181 180k	–
IN29	R182 DNP	R183 1.2M	–	R184 150k	R185 1.3M	R186 DNP	R187 DNP	–
IN30	–	R188 200k	R189 DNP	R190 130k	R191 DNP	R192 DNP	R193 DNP	–
IN31	R194 DNP	R195 DNP	R196 3M	R197 DNP	R198 DNP	–	R199 DNP	–

DC1c · weights IN32–IN63 (right bank) · neurons N8–N11

value = solder 5% 0805. **value** with **1%** tag on blue = solder 1% precision part. **DNP** = leave pads OPEN (never 0Ω). – on grey = no footprint on the board.

IN	POS N8 (POS_N0)	INV N8 (INV_N0)	POS N9 (POS_N1)	INV N9 (INV_N1)	POS N10 (POS_N2)	INV N10 (INV_N2)	POS N11 (POS_N3)	INV N11 (INV_N3)
IN32	R200 **110k**	–	R201 **91k**	R202 DNP	R203 DNP	R204 **47k**	R205 **910k**	R206 DNP
IN33	R207 **430k**	R208 DNP	R209 **270k**	–	R210 DNP	R211 **330k**	R212 DNP	–
IN34	R213 DNP	R214 DNP	R215 **160k**	–	R216 **910k**	R217 DNP	R218 DNP	R219 DNP
IN35	R220 DNP	R221 **390k**	R222 DNP	R223 DNP	R224 DNP	R225 **750k**	–	R226 DNP
IN36	R227 DNP	R228 **750k**	–	R229 **1.1M**	R230 DNP	R231 **1M**	R232 DNP	R233 DNP
IN37	R234 **270k**	R235 DNP	R236 DNP	R237 DNP	R238 **270k**	R239 DNP	–	R240 DNP
IN38	R241 **91k**	–	R242 DNP	R243 **510k**	R244 **510k**	R245 DNP	R246 DNP	–
IN39	R247 **240k**	R248 DNP	R249 **130k**	R250 DNP	R251 **270k**	R252 DNP	R253 DNP	R254 DNP
IN40	R255 DNP	R256 DNP	R257 **68k**	–	R258 DNP	R259 DNP	–	R260 DNP
IN41	R261 **240k**	R262 DNP	R263 DNP	–	R264 **390k**	–	–	R265 DNP
IN42	R266 **470k**	R267 DNP	R268 **300k**	–	R269 DNP	–	R270 DNP	–
IN43	R271 **300k**	R272 DNP	R273 **180k**	–	R274 **560k**	R275 DNP	–	R276 DNP
IN44	R277 **270k**	R278 DNP	R279 **270k**	–	R280 DNP	R281 DNP	–	R282 DNP
IN45	R283 **180k**	R284 DNP	R285 **270k**	R286 DNP	R287 **2M**	R288 DNP	–	R289 **240k**
IN46	R290 **150k**	–	R291 **270k**	–	R292 **430k**	R293 DNP	R294 DNP	R295 DNP
IN47	R296 **560k**	R297 DNP	R298 **39k**	R299 DNP	R300 DNP	R301 **360k**	R302 DNP	R303 DNP
IN48	R304 **1.1M**	R305 DNP	–	R306 DNP	R307 DNP	R308 **270k**	R309 DNP	–
IN49	R310 DNP	R311 DNP	R312 **270k**	–	R313 DNP	–	–	R314 DNP
IN50	R315 **220k**	R316 DNP	–	–	R317 DNP	R318 DNP	–	R319 DNP
IN51	R320 **220k**	R321 DNP	R322 **750k**	R323 DNP	R324 DNP	R325 DNP	R326 DNP	R327 DNP
IN52	R328 **620k**	R329 DNP	R330 DNP	R331 **750k**	R332 DNP	–	R333 DNP	R334 DNP
IN53	R335 **620k**	R336 DNP	R337 DNP	R338 **820k**	R339 DNP	R340 **620k**	–	R341 **300k**
IN54	–	R342 DNP	–	R343 **820k**	R344 DNP	R345 **200k**	R346 DNP	–
IN55	R347 **1.2M**	R348 DNP	R349 **56k**	R350 DNP	R351 **1.3M**	R352 DNP	R353 DNP	R354 DNP
IN56	–	R355 **62k**	R356 **110k**	R357 DNP	R358 DNP	R359 **100k**	–	R360 DNP
IN57	R361 **300k**	R362 DNP	R363 **130k**	R364 DNP	R365 DNP	R366 **330k**	–	R367 **150k**
IN58	–	R368 **680k**	R369 **220k**	R370 DNP	–	R371 **220k**	R372 **560k**	R373 DNP
IN59	–	R374 **360k**	R375 **390k**	–	–	R376 **270k**	R377 **300k**	R378 DNP
IN60	–	R379 **510k**	R380 **470k**	R381 DNP	R382 **820k**	R383 DNP	R384 **110k**	R385 DNP
IN61	R386 DNP	R387 **300k**	R388 **68k**	R389 DNP	R390 DNP	R391 DNP	–	R392 DNP
IN62	–	R393 DNP	R394 **110k**	R395 DNP	R396 **200k**	–	–	R397 **150k**
IN63	–	R398 **240k**	R399 **160k**	R400 DNP	R401 **36k**	R402 DNP	R403 DNP	R404 **560k**

DC1c · central resistors (feedback, bias, balance)

x = 47 / 52.6 mm, y = 59–94 mm. Balance and feedback are 1% (blue); bias is 5%. Red rows stay open.

Ref	X mm	Y mm	Pad 1	Pad 2	Role	Neuron	Value	Tol	Original ref
R405	47	59	+5V_REF	INV_N0	bias	N8	**820k**	5%	Rb_N8
R406	52.6	59	+5V_REF	INV_N1	bias	N9	**270k**	5%	Rb_N9
R407	47	64	+5V_REF	INV_N2	bias	N10	DNP (open)		
R408	52.6	64	+5V_REF	INV_N3	bias	N11	DNP (open)		
R409	47	69	+5V_REF	POS_N2	bias	N10	DNP (open)		
R410	52.6	69	GND	INV_N1	balance	N9	**18.2k**	1%	R_BAL_1_9
R411	47	74	GND	INV_N2	balance	N10	DNP (open)		
R412	52.6	74	GND	INV_N3	balance	N11	DNP (open)		
R413	47	79	GND	POS_N0	balance	N8	**8.66k**	1%	R_BAL_1_8
R414	52.6	79	GND	POS_N1	balance	N9	DNP (open)		
R415	47	84	GND	POS_N2	balance	N10	**35.7k**	1%	R_BAL_1_10
R416	52.6	84	GND	POS_N3	balance	N11	**90.9k**	1%	R_BAL_1_11
R417	47	89	INV_N0	V_SUM_0	feedback	N8	**100k**	1%	Rf_N8
R418	52.6	89	INV_N1	V_SUM_1	feedback	N9	**100k**	1%	Rf_N9
R419	47	94	INV_N2	V_SUM_2	feedback	N10	**100k**	1%	Rf_N10
R420	52.6	94	INV_N3	V_SUM_3	feedback	N11	**100k**	1%	Rf_N11

Also fit the fixed parts: U1 TL074, C_U1P and C_U1N 0.1 µF 50 V 0805, J_STRADDLE_EDGE.

DC1d · weights IN0–IN31 (left bank) · neurons N12–N15

value = solder 5% 0805. **value** with **1%** tag on blue = solder 1% precision part. **DNP** = leave pads OPEN (never 0Ω). – on grey = no footprint on the board.

IN	POS N12 (POS_N0)	INV N12 (INV_N0)	POS N13 (POS_N1)	INV N13 (INV_N1)	POS N14 (POS_N2)	INV N14 (INV_N2)	POS N15 (POS_N3)	INV N15 (INV_N3)
IN0	R001 **43k**	R002 DNP	R003 DNP	R004 DNP	R005 **39k**	R006 DNP	R007 DNP	R008 DNP
IN1	R009 **200k**	R010 DNP	–	R011 DNP	–	R012 **91k**	R013 DNP	R014 **220k**
IN2	R015 **390k**	R016 DNP	–	R017 DNP	R018 DNP	R019 **100k**	R020 DNP	–
IN3	R021 **200k**	R022 DNP	R023 DNP	R024 DNP	R025 DNP	R026 **150k**	R027 DNP	–
IN4	R028 **300k**	R029 DNP	R030 DNP	R031 DNP	–	R032 DNP	R033 DNP	–
IN5	R034 **270k**	R035 DNP	R036 DNP	R037 DNP	R038 DNP	R039 **200k**	R040 DNP	–
IN6	R041 **750k**	R042 DNP	–	R043 DNP	R044 DNP	R045 **110k**	R046 DNP	–
IN7	–	R047 **270k**	–	R048 DNP	R049 DNP	R050 DNP	R051 DNP	R052 DNP
IN8	R053 DNP	R054 **160k**	R055 DNP	R056 **430k**	–	R057 **36k**	R058 DNP	R059 **180k**
IN9	R060 **910k**	R061 DNP	R062 DNP	R063 DNP	R064 DNP	R065 **270k**	R066 DNP	–
IN10	R067 **300k**	R068 DNP	–	R069 DNP	–	R070 **360k**	–	R071 DNP
IN11	R072 **270k**	R073 DNP	R074 DNP	R075 DNP	R076 DNP	R077 **330k**	–	–
IN12	R078 **360k**	R079 DNP	R080 **750k**	–	–	R081 **1.1M**	–	R082 **220k**
IN13	R083 **390k**	R084 DNP	–	R085 DNP	–	R086 **270k**	R087 DNP	R088 DNP
IN14	R089 **330k**	R090 DNP	R091 DNP	R092 DNP	–	R093 **300k**	–	R094 DNP
IN15	–	R095 **220k**	–	R096 DNP	–	R097 **120k**	–	R098 DNP
IN16	R099 **330k**	R100 DNP	R101 DNP	R102 DNP	R103 **75k**	R104 DNP	R105 DNP	R106 DNP
IN17	R107 **510k**	R108 DNP	R109 DNP	R110 DNP	R111 **220k**	R112 DNP	R113 DNP	R114 DNP
IN18	R115 **1M**	R116 DNP	–	R117 DNP	R118 DNP	R119 **750k**	R120 DNP	R121 DNP
IN19	R122 **130k**	–	R123 DNP	R124 DNP	R125 DNP	R126 DNP	–	R127 DNP
IN20	R128 **110k**	–	–	R129 DNP	R130 DNP	R131 **820k**	–	R132 DNP
IN21	–	R133 DNP	–	R134 DNP	R135 DNP	R136 **2M**	–	R137 DNP
IN22	–	R138 **1.5M**	–	R139 DNP	R140 DNP	R141 DNP	–	R142 DNP
IN23	R143 **510k**	R144 DNP	–	R145 DNP	R146 **51k**	R147 DNP	–	R148 DNP
IN24	–	R149 **220k**	R150 DNP	R151 DNP	R152 DNP	R153 **150k**	R154 DNP	R155 DNP
IN25	R156 DNP	R157 **390k**	R158 DNP	R159 DNP	R160 DNP	R161 **160k**	R162 DNP	R163 DNP
IN26	R164 DNP	R165 **360k**	–	R166 DNP	–	R167 **150k**	R168 DNP	R169 DNP
IN27	R170 DNP	R171 **360k**	–	R172 DNP	–	R173 **1.3M**	R174 DNP	R175 DNP
IN28	R176 **560k**	R177 DNP	–	R178 DNP	R179 DNP	R180 **430k**	R181 DNP	–
IN29	R182 **330k**	R183 DNP	–	R184 DNP	R185 DNP	R186 **470k**	R187 DNP	–
IN30	–	R188 DNP	R189 DNP	R190 DNP	R191 DNP	R192 DNP	R193 DNP	–
IN31	R194 **240k**	R195 DNP	R196 DNP	R197 DNP	R198 **39k**	–	R199 DNP	–

DC1d · weights IN32–IN63 (right bank) · neurons N12–N15

value = solder 5% 0805. **value** with **1%** tag on blue = solder 1% precision part. **DNP** = leave pads OPEN (never 0Ω). – on grey = no footprint on the board.

IN	POS N12 (POS_N0)	INV N12 (INV_N0)	POS N13 (POS_N1)	INV N13 (INV_N1)	POS N14 (POS_N2)	INV N14 (INV_N2)	POS N15 (POS_N3)	INV N15 (INV_N3)
IN32	R200 **270k**	–	R201 DNP	R202 DNP	R203 DNP	R204 **470k**	R205 DNP	R206 DNP
IN33	R207 DNP	R208 **330k**	R209 DNP	–	R210 DNP	R211 **160k**	R212 DNP	–
IN34	R213 DNP	R214 **220k**	R215 DNP	–	R216 DNP	R217 **300k**	R218 DNP	R219 DNP
IN35	R220 DNP	R221 **180k**	R222 **130k**	R223 DNP	R224 DNP	R225 **360k**	–	R226 DNP
IN36	R227 DNP	R228 **330k**	–	R229 DNP	R230 DNP	R231 **160k**	R232 **160k**	R233 DNP
IN37	R234 **430k**	R235 DNP	R236 DNP	R237 DNP	R238 DNP	R239 **270k**	–	R240 DNP
IN38	R241 **220k**	–	R242 DNP	R243 DNP	R244 DNP	R245 **240k**	R246 DNP	–
IN39	R247 DNP	R248 **130k**	R249 DNP	R250 DNP	R251 **91k**	R252 DNP	R253 DNP	R254 DNP
IN40	R255 DNP	R256 **430k**	R257 DNP	–	R258 **680k**	R259 DNP	–	R260 DNP
IN41	R261 DNP	R262 **180k**	R263 DNP	–	R264 **160k**	–	–	R265 DNP
IN42	R266 DNP	R267 **470k**	R268 DNP	–	R269 **270k**	–	R270 DNP	–
IN43	R271 DNP	R272 **2.2M**	R273 DNP	–	R274 DNP	R275 **910k**	–	R276 DNP
IN44	R277 DNP	R278 **430k**	R279 DNP	–	R280 DNP	R281 **560k**	–	R282 DNP
IN45	R283 DNP	R284 **430k**	R285 DNP	R286 **180k**	R287 DNP	R288 **1.3M**	–	R289 DNP
IN46	R290 DNP	–	R291 DNP	–	R292 DNP	R293 DNP	R294 DNP	R295 DNP
IN47	R296 **120k**	R297 DNP	R298 DNP	R299 DNP	R300 **100k**	R301 DNP	R302 DNP	R303 DNP
IN48	R304 DNP	R305 **130k**	–	R306 DNP	R307 **150k**	R308 DNP	R309 DNP	–
IN49	R310 DNP	R311 **270k**	R312 DNP	–	R313 **68k**	–	–	R314 DNP
IN50	R315 DNP	R316 **300k**	–	–	R317 **82k**	R318 DNP	–	R319 **240k**
IN51	R320 DNP	R321 **200k**	R322 DNP	R323 DNP	R324 **62k**	R325 DNP	R326 DNP	R327 DNP
IN52	R328 DNP	R329 **150k**	R330 DNP	R331 **330k**	R332 **75k**	–	R333 **160k**	R334 DNP
IN53	R335 DNP	R336 **120k**	R337 DNP	R338 DNP	R339 **62k**	R340 DNP	–	R341 DNP
IN54	–	R342 **75k**	–	R343 DNP	R344 **68k**	R345 DNP	R346 DNP	–
IN55	R347 DNP	R348 **100k**	R349 DNP	R350 DNP	R351 **240k**	R352 DNP	R353 DNP	R354 DNP
IN56	–	R355 DNP	R356 DNP	R357 **200k**	R358 DNP	R359 **36k**	–	R360 DNP
IN57	R361 DNP	R362 **160k**	R363 DNP	R364 DNP	R365 DNP	R366 **82k**	–	R367 DNP
IN58	–	R368 **160k**	R369 DNP	R370 DNP	–	R371 **430k**	R372 DNP	R373 DNP
IN59	–	R374 **150k**	R375 **180k**	–	–	R376 DNP	R377 DNP	R378 DNP
IN60	–	R379 **1.2M**	R380 DNP	R381 DNP	R382 **180k**	R383 DNP	R384 DNP	R385 DNP
IN61	R386 DNP	R387 **160k**	R388 DNP	R389 DNP	R390 DNP	R391 DNP	–	R392 DNP
IN62	–	R393 **200k**	R394 DNP	R395 **200k**	R396 **330k**	–	–	R397 DNP
IN63	–	R398 **100k**	R399 DNP	R400 **390k**	R401 DNP	R402 **160k**	R403 DNP	R404 DNP

DC1d · central resistors (feedback, bias, balance)

x = 47 / 52.6 mm, y = 59–94 mm. Balance and feedback are 1% (blue); bias is 5%. Red rows stay open.

Ref	X mm	Y mm	Pad 1	Pad 2	Role	Neuron	Value	Tol	Original ref
R405	47	59	+5V_REF	INV_N0	bias	N12	**750k**	5%	Rb_N12
R406	52.6	59	+5V_REF	INV_N1	bias	N13	DNP (open)		
R407	47	64	+5V_REF	INV_N2	bias	N14	**200k**	5%	Rb_N14
R408	52.6	64	+5V_REF	INV_N3	bias	N15	DNP (open)		
R409	47	69	+5V_REF	POS_N2	bias	N14	DNP (open)		
R410	52.6	69	GND	INV_N1	balance	N13	DNP (open)		
R411	47	74	GND	INV_N2	balance	N14	**113k**	1%	R_BAL_1_14
R412	52.6	74	GND	INV_N3	balance	N15	DNP (open)		
R413	47	79	GND	POS_N0	balance	N12	**15.4k**	1%	R_BAL_1_12
R414	52.6	79	GND	POS_N1	balance	N13	**52.3k**	1%	R_BAL_1_13
R415	47	84	GND	POS_N2	balance	N14	DNP (open)		
R416	52.6	84	GND	POS_N3	balance	N15	**61.9k**	1%	R_BAL_1_15
R417	47	89	INV_N0	V_SUM_0	feedback	N12	**100k**	1%	Rf_N12
R418	52.6	89	INV_N1	V_SUM_1	feedback	N13	**100k**	1%	Rf_N13
R419	47	94	INV_N2	V_SUM_2	feedback	N14	**100k**	1%	Rf_N14
R420	52.6	94	INV_N3	V_SUM_3	feedback	N15	**100k**	1%	Rf_N15

Also fit the fixed parts: U1 TL074, C_U1P and C_U1N 0.1 µF 50 V 0805, J_STRADDLE_EDGE.